

Solutions

1. Let $f(x) = \sin(x)$, and let $c = 0$. Recall the n th order Taylor polynomial of f at c is

$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(c)}{k!} (x - c)^k.$$

- (a) Find $p_1(x)$, the first order Taylor polynomial of f at c . What is its degree?
(b) Find $p_2(x)$, the second order Taylor polynomial of f at c . What is its degree?

Solution. We compute $\sin$ and its derivatives at $c = 0$,

$$\sin(0) = 0, \quad \sin'(0) = \cos(0) = 1, \quad \sin''(0) = -\sin(0) = 0, \quad \sin^{(3)}(0) = -\cos(0) = -1,$$

and then write

$$\begin{aligned} p_1(x) &= \sin(0) + \sin'(0)x = x, \\ p_2(x) &= \sin(0) + \sin'(0)x + \frac{1}{2}\sin''(0)x^2 = x. \end{aligned}$$

The degree of both polynomials is 1.

2. Consider the forward difference approximation to the derivative

$$f'(c) \approx \frac{f(c+h) - f(c)}{h} =: F.$$

- (a) In general, if h is halved, what is expected to happen to the error $|f'(c) - F|$?
(b) What happens in the specific case of $f(x) = \sin(x)$ and $c = 0$? (Hint: use Taylor's Theorem with p_2 from question 1).

Solution. The error in the forward difference approximation to the derivative is given by the expression

$$f'(c) - \frac{f(c+h) - f(c)}{h} = -\frac{f''(\xi)}{2}h,$$

for some ξ between c and $c+h$. The right-hand side scales like h , so if h is halved, we expect the error to be roughly halved as well. The error will not generally be *exactly* halved, because we evaluate f'' at a different ξ .

In the specific case of $f(x) = \sin(x)$ and $c = 0$, note that $f''(\xi) = -\sin(\xi)$, and for ξ close to 0, $\sin(\xi) \approx \xi$, so this suggests the error may actually be smaller than predicted above. We can make this rigorous using $p_2(x)$. Taylor's Theorem with h gives

$$\sin(h) = p_2(h) + \frac{\sin'''(\xi)}{3!}h^3 = h - \frac{\cos(\xi)}{6}h^3.$$

The forward difference quotient is

$$\begin{aligned} \frac{f(c+h) - f(c)}{h} &= \frac{\sin(h) - \sin(0)}{h} = \frac{h - \frac{1}{6}\cos(\xi)h^3}{h} \\ &= 1 - \frac{1}{6}\cos(\xi)h^2 = \sin'(0) - \frac{1}{6}\cos(\xi)h^2, \end{aligned}$$

so, because the first and second Taylor polynomials are the same, we get an extra cancellation, and in this particular case, if we halve h , we expect the error to decrease by a factor of about 4.